Lab 2 - Final Task

Extracted from the final mini-project/task section at the end of the notebook.

Source notebook: Lab2.ipynb **Extracted cell:** 54

17. Final mini-project

Complete this part alone in a new notebook. Do not copy the solution notebook while working.

Use the same Wine dataset and the helper function `make_dirty_wine_dataset()`.

Your mini-project must include the following:

1. Load the Wine dataset into a DataFrame.
2. Create the dirty teaching copy using `make_dirty_wine_dataset()`.
3. Display the first 10 rows, shape, columns, missing values, duplicate count, and descriptive statistics.
4. Identify the target column, identifier column, numerical columns, and categorical columns.
5. Create at least two missing-indicator columns.
6. Remove duplicate rows.
7. Convert invalid negative values in positive-only numerical columns to missing values.
8. Clean inconsistent category labels in `alcohol_level`.
9. Impute numerical missing values using the median and categorical missing values using the mode.
10. Create at least two new derived columns.
11. Use `groupby()` to compare at least two features by wine class.
12. Create one missing-value bar chart and one boxplot.
13. Create one Seaborn heatmap after cleaning.
14. Build a leakage-safe preprocessing pipeline using train-test split and `ColumnTransformer`.
15. Write two short paragraphs:

• one about the most important data-quality problem you found;

• one about why safe preprocessing is important before future AI models.